Procedural Adaptive Footstep System – AUD Task Plan (Latest)

This document summarizes the current status of the Procedural Adaptive Footstep System, including completed milestones and the next planned development tasks.

EPIC 1 – Core Architecture & Audio Foundations (COMPLETED)

- AUD-01: JUCE project setup and audio callback lifecycle
- AUD-02: Thread-safe parameter bridge using std::atomic
- AUD-03: Sample rate and block size management

EPIC 2 – Footstep Timing & Adaptation (COMPLETED)

- AUD-04: Speed-based gait detection (Sneak / Walk / Run)
- AUD-05: Phase accumulator and deterministic step triggering
- AUD-06: Dead-zone handling and movement validation

EPIC 3 – Procedural Sound Generation (COMPLETED)

- AUD-07: Procedural impact noise generation
- AUD-08: Envelope shaping with weight-based decay
- AUD-09: Cloth noise layer integration

EPIC 4 – Surface & Material Awareness (PARTIALLY COMPLETED)

- AUD-10: Surface parameter plumbing via atomic enum index
- AUD-11: Surface-to-DSP mapping (brightness / filtering) – IN PROGRESS

EPIC 5 – Voice Management & Polyphony (COMPLETED)

- AUD-12: Fixed-size FootstepVoice pool
- AUD-13: Voice allocation and deterministic voice stealing
- AUD-14: Overlapping footstep playback without artifacts

EPIC 6 – System Stability & Safety (COMPLETED)

- AUD-15: Audio-thread safety audit (no allocations, no locks)
- AUD-16: Proper voice lifecycle handling (isActive logic)

EPIC 7 – UI & Debug Tools (NEXT TASK)

- AUD-17: Debug UI sliders for speed, weight, cloth intensity
- AUD-18: Surface selector (ComboBox) for runtime testing
- AUD-19: Visual feedback for gait, step triggers, and active voices
- AUD-20: Validation of real-time parameter response with no audio glitches

EPIC 8 – Future Work (NOT STARTED)

- Surface-dependent spectral shaping
- Left / right foot alternation
- Gameplay and animation event hooks
- Profiling, stress testing, and portfolio documentation